MIL-E-1/1091B
18 August 1971
SUPERSEDING
MIL-E-1/1091A(NAVY)
12 July 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12AU6

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, miniature, rf, sharp cutoff

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5 1/2

Cathode --- Coated unipotential

Base connections:

Pin No.		1	2	3	4	5	6	7
Element	---	g1	g3 int sd	h	h	a	g2	k

ABSOLUTE MAXIMUM RATINGS:

	(B)								(B)
Parameter:	Ef	Eb	Ec1	Ec2	Pg2	Pp	Ehk	Rk	Alt
Unit:	V	Vdc	Vdc	Vdc	W	W	v	Ohms	ft
Maximum:	13.9	330	---	165	0.7	3.3	100	---	Note 1
Minimum:	11.3	---	---	---	---	---	---	---	---
TEST CONDITIONS:	12.6	250	0	150	---	---	---	68	---

GENERAL:

Qualification - Required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Ec1 = -1.5 Vdc (see note 2)	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	8.0	13.5	mAdc
1306	Transconductance		0.65	II	Sm	4,150	6,250	µmhos
(B) 1231	Emission	Eb = Ec1 = Ec2 = 20 Vdc (see note 2)	0.65	II	Is	60	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	138	162	mA
(B) 1211	Insulation of electrodes		---	---	---	---	---	---
(B) 1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1256	Electrode current (2) (anode)	Ec1 = -8 Vdc	---	---	Ib	----	50	µAdc
1256	Electrode current (screen)		---	---	Ic2	2.6	6.0	mAdc
1331	Direct-interelectrode capacitance	With shield No. 316	---	---	Cgp	---	0.0035	pF
					Cin	4.4	6.6	pF
					Cout	3.5	6.5	pF
(B) 1121	Base strain			---	---	---	---	---
(B) 2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
(B) 1501	Intermittent life	Group A: Ehk = 100 V; Rg1 = 0.05 Meg (min)	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)	Transconductance	---	---	Sm	3,400	---	µmhos

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2553)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVAL ELECTRONIC SYSTEMS COMMAND
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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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